

Metal-Free, Low-Surface Energy, and Self-Healing Polyurethane Coating with an Excellent Antifouling Property

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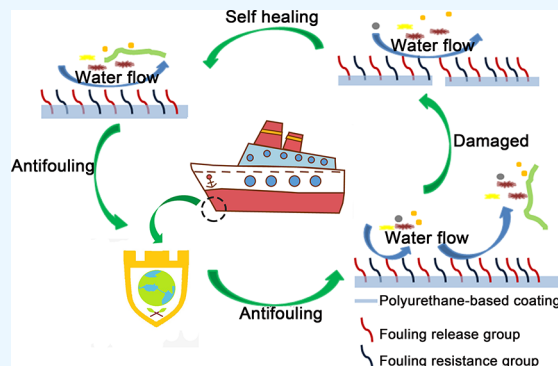
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ABSTRACT: The prevention and control of water biofouling has always been the focus of antifouling research. Coatings containing metal ions can effectively inhibit biofouling. However, the application of this type of coating has been limited as a result of its toxicity, irreparability, and poor antifouling ability under static conditions. Here, we have synthesized a self-repairing polymer antifouling coating with hydrophobic and multiple self-repairing systems through structural design with polytetrahydrofuran (PTMEG), isophorone diisocyanate (IPDI), perfluorooctyl ethyl acrylate (TEAc-8), triclosan (TCS), and 3,3-dithiodipropionic acid as a raw material. The prepared coating can realize a self-repairing function at room temperature in both air and water, thereby prolonging the antifouling life of the coating. A large number of hydrogen bonds in the molecular structure enable the coating to have good adhesion to the substrate. At the same time, the lower surface energy (<20 mN/m) and higher hydrophobic angle ($>110^\circ$) of the coating determine the material's good fouling release ability. In addition, the lake antifouling experiments show that the coating has good static antifouling ability. The coatings prepared will provide an alternative strategy for designing self-healing marine antifouling coatings in practical applications.

KEYWORDS: environmentally friendly, polyurethane, fouling release, fouling resistance, self-healing



1. INTRODUCTION

Microorganisms, plants, and animals in the river or lake will attach, grow, and reproduce on equipment such as ships,¹ accelerating the corrosion of equipment, shortening the service period of the equipment, and causing serious economic and environmental problems.^{2–5} The formation of fouling on the substrate surface goes through four stages. The four stages are the formation of the conditioned film, biofilm, adhesion of small organisms, and adhesion of large organisms, which take one to two months in total.⁶

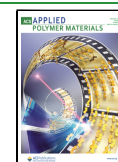
Among many antifouling methods, painting antifouling coatings is the most economical and effective method.^{7,8} Coatings containing tributyltin (TBT) were once the most effective system to prevent biofouling, but due to their persistent toxicity to non-target organisms, they have been globally banned since 2008.^{9,10} Subsequently, antifouling materials have gradually been developed in the direction of low toxicity and environmental protection. At present, environmentally friendly antifouling coatings include fouling release coatings, fouling resistance coatings, bionic antifouling coatings, nanocomposite antifouling coatings, and amphiphilic antifouling coatings.^{11–15} Fouling release coatings have become a research hotspot owing to their environmental protection and drag reduction properties.^{16,17}

Fouling release coatings mainly rely on their lower surface energy to prevent the adhesion of fouling organisms or remove fouling organisms from the surface through water washing and mechanical removal to achieve the effect of antifouling.^{18–26} However, many studies have proved that the fouling release coatings cannot resist biological adhesion in the absence of a strong water flow. The materials also have problems of poor mechanical properties and poor substrate adhesion, which result in the limited range of application.^{10,27–29} So far, people have made great efforts to improve the mechanical properties and antifouling performance of the fouling release coatings.^{30–40} For example, Liu et al.⁴⁰ prepared polydimethylsiloxane (PDMS)-based polyurea with a soft segment of PDMS and a hard segment of 1,6-ethylenediamine (HMDA). The urea group in the hard segment can form high-density hydrogen bonds with the substrate, thereby improving the adhesion of the coating. Xie et al.²⁹ used a chemical grafting method to fix the antifouling group on the PDMS elastomer,

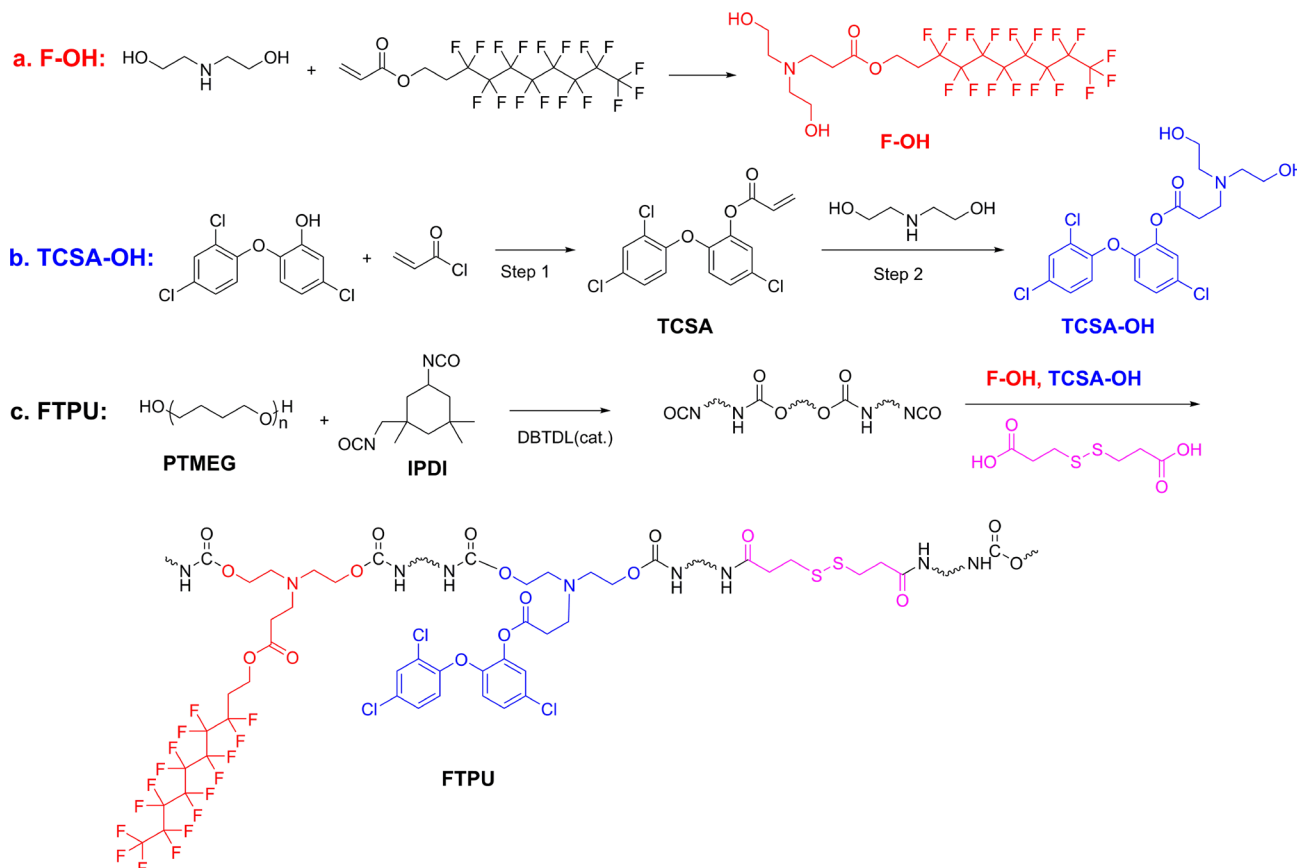
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Scheme 1. Synthesis of F-OH (a), TCSA-OH (b), and FTPU (c)



which not only effectively avoided pollution of the environment from the release of antifouling agents but also improved the static antifouling ability of the coating. However, these modification methods cannot simultaneously optimize mechanical properties and fouling release and fouling resistance performance. Recently, Chen et al.¹⁴ prepared an amphiphilic telomer by a sol-gel method and mixed it with triethoxyoctylsilane (KH832) and tetraethylorthosilicate (TEOS) to prepare an organic-inorganic hybrid coating, which has high toughness, adhesion strength to the substrate, and good static antifouling performance, pushing the development of antifouling coatings to a high level. However, long-term use of the coating will inevitably be damaging, thus shortening the antifouling life.

In order to extend the life of the antifouling coating, a coating matrix with self-healing properties is essential in practical applications. In fact, reversible associations such as disulfide bonds and hydrogen bonds can be combined to prepare a matrix with high-efficiency self-repair capabilities.^{41,42} For example, Xu et al.⁴³ prepared a poly(urea-polyurethane) (PUU)-based composite material with inherent room-temperature self-healing ability through hydrogen bonding in the PUU matrix. Therefore, by combining the excellent mechanical properties and good adhesion of polyurethane (PU) coatings, a self-healing polyurethane antifouling coating can be designed to inhibit freshwater biofouling in practical applications.

In this study, we prepared a type of environmentally friendly self-healing polyurethane antifouling coating (FTPU) grafted with triclosan diol (TCSA-OH) and fluorinated glycol (F-OH). Among them, triclosan is a spectrum of bactericide and

antialgae. The large number of hydrogen bonds in the polyurethane coating structure that we designed can effectively improve the mechanical properties of the material and the adhesion of the substrate. During the formation of the coating, the antifouling group was dragged to the surface of the coating owing to the migration of the fluorocarbon chain, which was beneficial for the coating to exert the effect of fouling resistance and achieve an excellent static antifouling effect. More importantly, the chemical fixation of the antifouling agent made the coating more durable and environmentally friendly. It was considered that the coating was susceptible to damage and the antifouling effect was ineffective. We had introduced disulfide bonds into the coating, which allowed polyurethane to realize the synergistic self-repair of hydrogen bonds and disulfide bonds under isothermal conditions at room temperature or external stimulation, thereby extending the service life of the coating. We tested and evaluated the mechanical properties and self-repairing properties of the materials. We also studied the antifouling performance by testing the surface energy of the coating. Through lake field experiments, the long-term antifouling effect was evaluated. Our research provides a method for preparing self-healing antifouling coatings in practical applications.

2. EXPERIMENT SECTION

2.1. Materials. Isophorone diisocyanate (IPDI), polytetrahydrofuran (PTMEG, $M_n = 2000$), 1,4-butanediol (BDO), 3,3'-dithiodipropionic acid (S-S), diethanolamine (DEA), triclosan (TCS), acrylic chloride (AC), trimethylamine (TEA), dibutyltin dilaurate (DBTDL), dichloromethane

(DCM), and *N,N*-dimethylethanolamine (DMAC) from Aladdin were used as received. Methanol (MT) and ethanol (EA) were obtained from Tianjin Kermel Chemical Reagent Development, and perfluorooctyl ethyl acrylate (TEAc-8) was acquired from Shang Fluoro Ltd. Co.

2.2. Synthesis of F-OH. F-OH was synthesized from TEAc-8 and DEA (Scheme 1a), and the structure was confirmed by FT-IR, ^1H NMR, and ^{19}F NMR. Typically, 5.29 g of TEAc-8 was dissolved in 5 mL of MT and then poured into a three-necked round-bottom flask. Then, 1.16 g of DEA was added into the flask. The mixture was stirred for 24 h at 55 °C.

2.3. Synthesis of TCSA. TCSA was synthesized from TCS and AC (Scheme 1b), and the structure was confirmed by FT-IR, ^1H NMR, and ^{13}C NMR. Six grams of TCS and 2.10 g of TEA was dissolved in 20 mL of DCM and then poured into a 100 mL three-necked flask with a constant-pressure funnel. An amount of 1.88 g of AC was dissolved in 10 mL of DCM, and the solution was added to the constant-pressure funnel. The AC solution was added dropwise under ice-bath conditions, the dropping was completed in approximately 15 min, and then it was reacted for 8 h.

2.4. Synthesis of TCSA-OH. TCSA-OH was synthesized from TCSA and DEA (Scheme 1b), and the structure was confirmed by FT-IR, ^1H NMR, and ^{13}C NMR. TCSA (3.44 g) was dissolved in 5 mL of MT and then poured into a three-necked round-bottom flask. Then, 1.58 g of DEA was added into the flask. The mixture was stirred for 16 h at 55 °C.

2.5. Synthesis of the Self-Healing Polyurethane-Based Antifouling Coating. FTPU was synthesized from PTMEG, IPDI, F-OH, TCSA-OH, and S-S (Scheme 1c), and the structure was confirmed by FT-IR. Twenty grams of PTMEG, 8.89 g of IPDI, and 0.25 g of DBTDL were added into a three-necked flask connected with an electric stirrer and a condenser, stirred evenly under a N_2 atmosphere, and then placed in an oil bath at 75 °C for 3 h. F-OH (6.23 g), which accounts for 1/4 of the IPDI substance (the control group PU and TPU use BDO in the same amount instead of F-OH), was dissolved in 15 mL of DMAC, and the solution was added into the flask and reacted for 3.5 h. Then, 4.48 g of TCSA-OH, which accounts for 1/4 of the IPDI substance (the control group PU and FPU use BDO in the same amount instead of TCSA-OH), was dissolved in 15 mL of DMAC, and the solution was added into the flask and reacted for 2.5 h. Finally, we added 2.10 g of 3,3-dithiodipropionic acid, which accounts for 1/4 of the IPDI substance, and continued the reaction for 3 h at 85 °C. The reaction was monitored in FTIR. When the band at 2270 cm^{-1} for isocyanate completely disappeared, the reaction mixtures were dried under vacuum for 24 h at 100 °C.

2.6. Characterization. FT-IR spectra were recorded with Fourier transform infrared spectroscopy (Bruker, EQUINOX 55). The spectra were collected by a KBr disk method, and the scanning range was 400–4000 cm^{-1} with a spectral resolution of 4 cm^{-1} . All ^1H NMR, ^{19}F NMR, and ^{13}C NMR spectra were recorded on a Bruker AV600 NMR spectrometer using deuterated chloroform (CDCl_3) as solvent. Thermogravimetric analysis of the samples was done with an STA 449 F3 Jupiter thermal weight-loss analyzer produced by NETZSCH in Germany. We placed a certain quality sample in a special aluminum vessel and heated it at a heating rate of 10 °C/min from 30 to 800 °C in a nitrogen atmosphere. X-ray photoelectron spectroscopy of the samples was performed by an Axis Ultra DLD (Kratos Analytical) spectrometer. The

instrument uses an Al $K\alpha$ ($h\nu = 1486.6$ eV) ray source. The incident angle of the instrument is 90°. The pressure is 5×10^{-9} Torr. The voltage is 15 kV, and the current is 5 mA.

3. RESULTS AND DISCUSSION

Scheme 1 shows the preparation process of anti-fouling materials F-OH (a), TCSA-OH (b), and the self-healing polyurethane anti-fouling coating FTPU(c). In order to keep the overall structure of the polyurethane design unchanged and to study the effect of adding different antifouling systems on the antifouling effect of the polyurethane coating, we prepared four types of polyurethane samples, namely, pure polyurethane PU without the antifouling system, FPU with the fouling release system, TPU with the fouling resistance system, and FTPU with the integrated fouling release and fouling resistance system. The content of raw materials used in different polyurethane-based antifouling coatings is shown in Table 1.

Table 1. Formulation for the Synthesis of the Self-Healing Polyurethane-Based Antifouling Coating

sample	IPDI (m/g)	PTMEG2000 (m/g)	F-OH (m/g)	BDO (m/g)	TCSA- OH (m/g)	S-S (m/g)
PU	8.89	20	0	1.80	0	2.10
FPU	8.89	20	6.23	0.90	0	2.10
TPU	8.89	20	0	0.90	4.48	2.10
FTPU	8.89	20	6.23	0	4.48	2.10

3.1. FT-IR and NMR Characterization of F-OH and TCSA-OH. In order to characterize the structural characteristics of synthetic raw materials, FT-IR and NMR characterization were performed on F-OH and TCSA-OH (Figures S1–S8). The results showed the successful synthesis of F-OH and TCSA-OH.

3.2. FT-IR Characterization of PU, FPU, TPU, and FTPU. In order to characterize the structure of the synthesized sample, FT-IR characterization was performed on PU, FPU, TPU, and FTPU, and the results are shown in Figure S9. The bands at 3285 cm^{-1} can be ascribed to the $-\text{NH}-$ bonds.⁴⁴ The absorption peaks at about 2931 and 2856 cm^{-1} represent the $-\text{CH}_2-$ stretching mode.⁴⁵ The absorption peak at 2264 cm^{-1} corresponding to the $-\text{NCO}$ stretched band completely disappeared, indicating the complete consumption of $-\text{NCO}$ groups in the reaction and the successful preparation of polyurethane.⁴⁶ The bands at 1715 cm^{-1} can be ascribed to the vibration of $\text{C}=\text{O}$ in $-\text{COO}-$ and $-\text{NHCOO}-$ groups.^{47–49} The absorption peaks at about 1633 cm^{-1} represent the asymmetric $-\text{COO}-$ stretching mode and the $-\text{NH}-$ bending mode.⁵⁰ The bands at 1498 cm^{-1} can be ascribed to the $\text{C}=\text{C}$ stretching of the benzene ring.⁵¹ The absorption peaks at about 1393 cm^{-1} represent the vibration of $\text{C}-\text{O}$ in $-\text{COO}-$ and $-\text{NHCOO}-$ groups.⁵² The bands at 1186, 1103, and 1006 cm^{-1} can be assigned to the $\text{C}-\text{O}-\text{C}$ stretching for urethane.⁵³ The bands at 586 cm^{-1} can be ascribed to the $-\text{S}-\text{S}-$ stretching mode. Therefore, the FT-IR spectroscopic results indicated that the self-healing polyurethane antifouling coatings have been successfully synthesized.

3.3. Thermogravimetric Analysis (TG). In order to characterize the thermal stability of PU, FPU, TPU, and FTPU, the samples were tested by thermogravimetric analysis. The results are shown in Figure S10. It can be seen from Figure S10a that the thermal stability of FPU, TPU, and FTPU

under 210 °C was better than that of PU. The introduction of F-OH and TCSA-OH improved the coatings' early heat resistance as a result of the high bond energy of the C–F bond in F-OH and the good thermal stability of the benzene ring in TCSA-OH. As the temperature increases, the small molecules of F-OH and TCSA-OH on the polyurethane branch were decomposed, so the weight loss ratio of FPU, TPU, and FTPU increased.

As shown in Figure S10b, the four samples have three identical stages of weight loss in the temperature range of 200–420 °C, and the degradation mechanism is mainly affected by the chemical structure of the component with the lowest bond energy. The initial weight loss of the sample occurred at 210–250 °C, which corresponds to the decomposition of the amide bond. The second weight loss of the sample occurred at about 260–360 °C, which corresponds to the decomposition of the carbamate bond, and the weight loss around 380–420 °C is the characteristic of the degradation of the soft segment.

3.4. X-ray Photoelectron Spectroscopy (XPS). Experiments have proved that the distribution of antifouling groups on the surface is very important for the antifouling of polyurethane coatings. We used XPS to study the composition of the coating surface of PU, FPU, TPU, and FTPU, and the results are shown in Figure S11. As can be seen from Figure S11, there are strong 1s peaks of the C element, O element, and N element in the spectrum of each sample since the four samples are polyurethane-based coatings. Due to the surface migration ability of the F element, the peak of the F element appears in the FPU and FTPU spectra, indicating that the antifouling group F-OH exists on the surface of the coating, which is beneficial to exerting the fouling release effect. The distribution of the Cl element can be observed in the TPU and FTPU spectra, indicating that the coating surface has the distribution of the antifouling group TCSA-OH. In addition, it is found from the high-resolution Cl spectrum that the Cl element of FTPU has a stronger response. This shows that, during the curing process of FTPU, more TCSA-OH antifouling groups will be dragged to the surface of the coating due to the migration of the fluorocarbon chain, which is beneficial to the coating to exert the fouling resistance effect and obtain a better static antifouling effect.

3.5. Water Contact Angle and Surface Energy. As shown in Figure 1a, we studied the surface energy of polyurethane-based antifouling coatings. In general, polyurethane-based antifouling coatings with antifouling systems all exhibit low-surface energy characteristics (<25 mN/m). The surface energy of pure polyurethane PU is 27.15 mN/m. The

surface energy of FPU is as low as 14.86 mN/m as a result of the low surface energy of fluorine. TPU also has certain low-surface energy characteristics owing to TCSA-OH containing a benzene ring, which has a certain degree of hydrophobicity, and its surface energy is 22.16 mN/m. Among them, FTPU with the addition of F-OH and TCSA-OH has the lowest surface energy of 12.45 mN/m. This shows that the simultaneous introduction of F-OH and TCSA-OH can make the coating have a lower surface energy, which is conducive to the desorption of fouling organisms from the surface of the polyurethane-based coating.

As shown in Figure 1b, the water contact angles of PU, FPU, TPU, and FTPU are 92.9°, 113.5°, 101.5°, and 117.8°, respectively, and this is consistent with the test results of surface energy in Figure 1a. The contact angle of each sample changed a little after the coating was immersed in lake water. After three months of soaking, the contact angle only decreased by approximately 5°, indicating that the coating has long-term and stable hydrophobicity. The antifouling group can be concentrated on the surface of the coating for a long time, which is beneficial to maintaining the long-term effect of its fouling release ability.

3.6. Adhesion Tests. The main component of glass is silica, which is a hydrophilic substrate. To test the adhesion of the coatings on different material surfaces, tinplate and glass were selected as substrates. According to the ISO standard of adhesion, we have classified the grades of the coatings, and the grades are shown in Table S1.

Figure 2a–d shows the peeling off of the four polyurethane-based coatings on the tinplate-based scribing, and e–h show

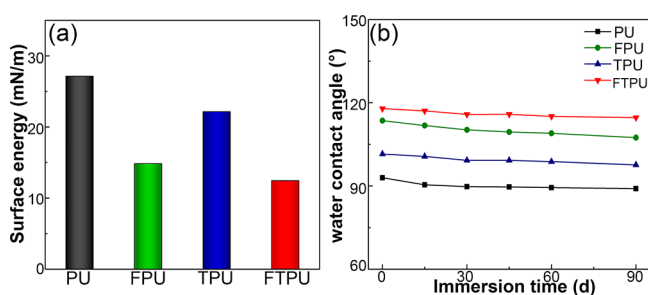


Figure 1. (a) Surface energy of polyurethane-based coatings. (b) Time dependence of the water contact angle of polyurethane-based coatings after immersion in lake water at 25 °C.

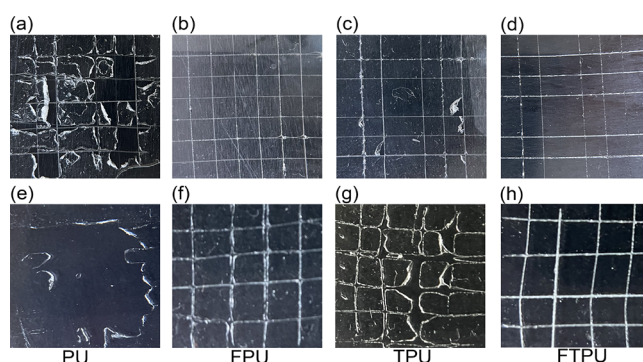


Figure 2. Peeling conditions of the coatings' cross-cut areas with tinplate as the base, namely, (a) PU, (b) FPU, (c) TPU, and (d) FTPU, and with the glass plate as the base: (e) PU, (f) FPU, (g) TPU, and (h) FTPU.

the four coatings peeling off on the glass-based scribing. It can be seen from the figure that PU showed signs of shedding on both substrates and the adhesion of the composite coating was higher than that of PU. In fact, the introduction of F-OH and TCSA-OH may increase the number of hydrogen bonds in the polyurethane system, and the increase of hydrogen bonds makes more –NHCOO– groups in contact with the substrate surface and improves the adhesion of the coating. Among them, FPU has good adhesion on both substrates, and the shedding area is not more than 5%. This is mainly because the introduction of fluorine increases the hydrogen bond in the coating and improves the adhesion to the substrate. It can be seen from Figure 2c,g that the TPU coating is less likely to fall off from the tinplate-based substrate, possibly owing to the fact

that the hydrogen bonds in the coating are more likely to bond with the tinplate surface. In Figure 2d,h, the cut edges of the FTPU coating are complete and smooth, and there is no peeling off on the edges of the grid. All in all, the polyurethane-based antifouling coating added with F-OH and TC-SA-OH has good adhesion to various substrates such as metal and glass, which is conducive to its long-term use in water environments.

3.7. Self-Healing Performance Research. **3.7.1. Intuitive and Microscopic Study of Self-Healing Performance.** We tested the self-repair performance of the sample FTPU under three conditions (Figures 3 and 4). As shown in Figure 3, the

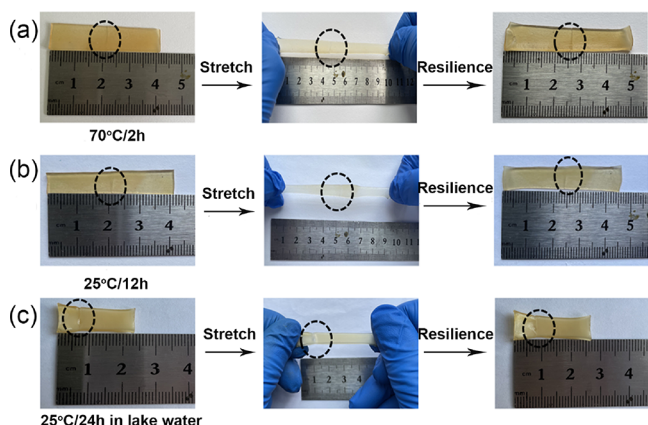


Figure 3. Intuitive test charts of stretch and resilience test of FTPU samples repaired under three repair conditions: (a) 70 °C/2 h, (b) 25 °C/12 h, and (c) 25 °C/12 h in lake water.

samples were completely cut from the middle with a scalpel and then assembled together carefully without external pressure. Then, they were repaired at 70 °C for 2 h then at room temperature of 25 °C for 12 h and then in lake water for 24 h. Finally, the repaired samples were subjected to tensile and rebound tests. The results show that FTPU has good self-

healing performance under the three test conditions, and the repaired samples also have a certain degree of resilience.

In order to analyze the repair performance of different degrees of coating damage from a microscopic point of view, a scalpel was used to scrape and cut the surface of the FTPU spline. The samples were then placed in an electric heating, constant-temperature blast, drying oven at 70 °C for 2 h then at room temperature of 25 °C for 12 h and then in lake water at 25 °C for 24 h. Finally, an electron microscope was used to observe the repair condition of the spline surface, and the result is shown in Figure 4. It can be seen from the figure that the samples can be repaired within a certain period of time whether it is thermally stimulated, at room temperature, or in lake water. The most important thing is that the sample FTPU can achieve self-repairing function in lake water, which is of great significance to the application of this material in biofouling.

3.7.2. Study on Mechanical Stretching of Samples Before and After Healing. In order to characterize the mechanical tensile properties of the samples, a mechanical tensile stress experiment was performed on PU, FPU, TPU, and FTPU, and the results are shown in Figure S12. It can be seen from the figure that the elongation of our synthesized linear polyurethane presents a parabolic trend. The introduction of F-OH and TC-SA-OH enhanced the tensile strength of polyurethane. The tensile strength of FTPU reached a maximum of 1.21 MPa, and the elongation at break was 449%. This is mainly due to the fact that F-OH and TC-SA-OH are added as hard segment materials into the polyurethane system, which enhances the degree of chemical cross-linking of molecular segments and the intermolecular force of polyurethane. At the same time, the introduction of fluorine atoms increases the number of hydrogen bonds in the polyurethane system, and the introduction of benzene rings increases the rigidity of the polyurethane chain, thereby further improving the tensile strength of the coating film.

We performed tensile tests on the samples before and after healing in order to further analyze the fracture repair

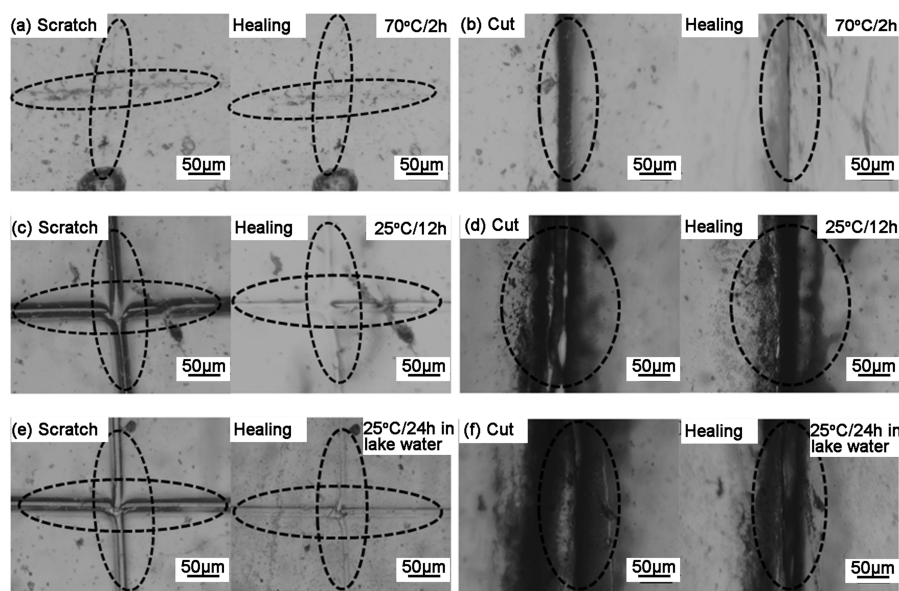


Figure 4. Microscopic test diagrams of the scratched and cut FTPU samples before and after healing under three repair conditions: (a, b) 70 °C/2 h, (c, d) 25 °C/12 h, and (e, f) 25 °C/12 h in lake water.

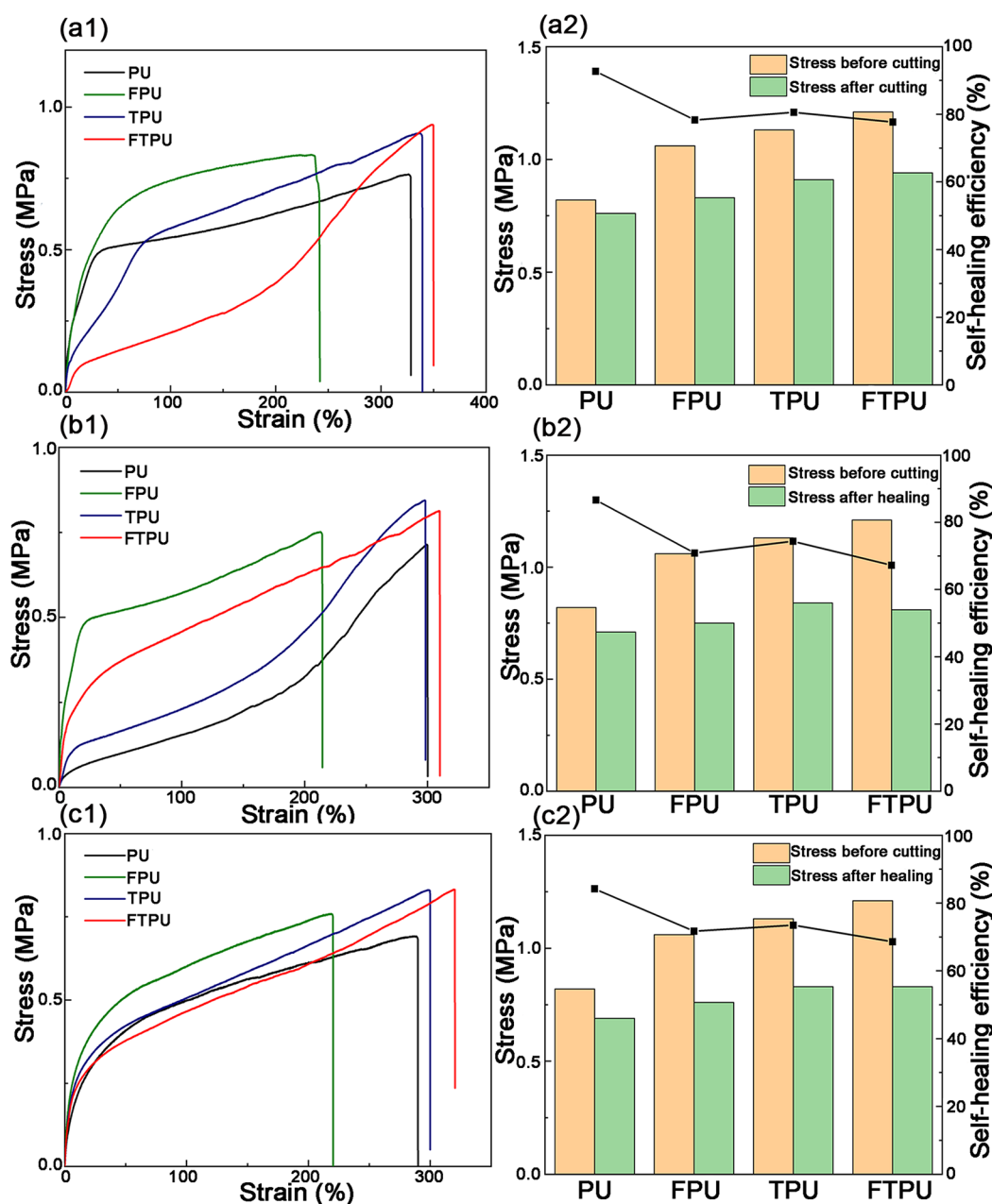


Figure 5. Strain–stress curves of full-cut polyurethane-based samples repaired under three repair conditions (a1 - 70 °C/2 h, b1 - 25 °C/12 h, and c1 - 25 °C/24 h in lake water) and comparison of the tensile strength of the sample before and after healing and its self-healing efficiency under three repair conditions (a2 - 70 °C/2 h, b2 - 25 °C/12 h, and c2 - 25 °C/24 h in lake water).

performance of the sample and study the self-healing efficiency under the three repair conditions based on the above microscopic research results on the crack repair of the spline. First, the samples were completely cut with a scalpel and repaired under three repair conditions [(a) 70 °C/2 h, (b) 25 °C/12 h, and (c) 25 °C/24 h in lake water], and then the repaired samples were subjected to a mechanical tensile stress test. The results are shown in Figure 5 and Table 2. It can be seen that the samples have different self-healing efficiencies under different repair conditions. Among them, all samples showed the best self-healing efficiency after repairing at 70 °C for 2 h. The repairing efficiencies of PU, FPU, TPU, and FTPU reached 92.68%, 78.30%, 80.53%, and 77.68%, respectively. This is because thermal stimulation may accelerate the movement and diffusion of molecular chains

on adjacent fracture surfaces to make the disulfide bonds close to each other and complete the reversible exchange reaction through the free radical-mediated mechanism so as to better realize the synergy of hydrogen bonds and disulfide bonds repairing. Finally, the repaired samples have good mechanical properties again. Also, the introduction of F-OH and TC-SA-OH reduced the self-healing efficiencies of polyurethane-based coatings as a result of the antifouling groups giving the polymer higher mechanical strength and limiting the mobility of the polymer to a certain extent at the same time. In fact, the self-repairing behavior of polymers is essentially the result of molecular chain movement and mutual diffusion and entanglement. It can be seen from Figure 5b2,c2 that the samples also have good self-healing efficiency at room temperature and in lake water. The repaired samples also

Table 2. Tensile Strength, Ultimate Elongation, and Healing Efficiency of the Samples under the Original State and Different Repair Conditions

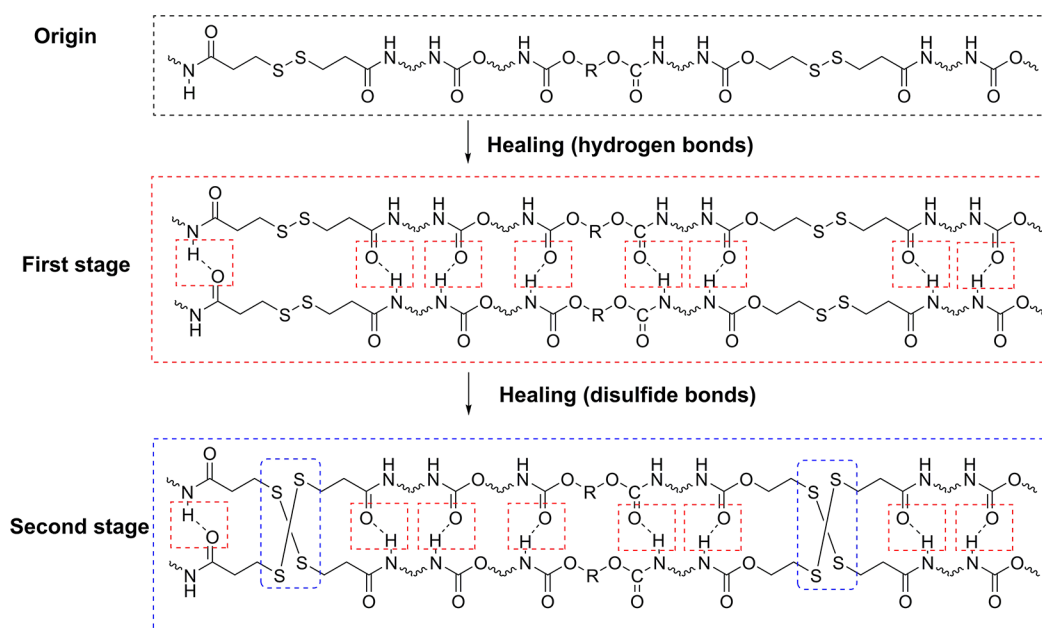
sample	repair condition	tensile strength (MPa)	ultimate elongation (%)	healing efficiency (%)
PU	before cutting	0.82	350	
	70 °C/2 h	0.76	327	92.68
	25 °C/12 h	0.71	299	86.59
	25 °C/24 h in water	0.69	287	84.15
FPU	before cutting	1.06	292	
	70 °C/2 h	0.83	242	78.30
	25 °C/12 h	0.75	214	70.75
	25 °C/24 h in water	0.76	222	71.70
TPU	before cutting	1.13	410	
	70 °C/2 h	0.91	337	80.53
	25 °C/12 h	0.84	298	74.34
	25 °C/24 h in water	0.83	304	73.45
FTP	before cutting	1.21	414	
	70 °C/2 h	0.94	350	77.68
	25 °C/12 h	0.81	312	66.94
	25 °C/24 h in water	0.83	322	68.59

have good mechanical properties. Among the samples, the self-healing efficiency of FTPU can reach 66.94% and 68.59% at room temperature and in lake water. The tensile strengths of the final repaired samples were 0.81 and 0.83 MPa. This means that this rapid self-healing process can occur in water and air, making the material better for antifouling.

3.7.3. Self-Healing Mechanism of Self-Healing Polyurethane Containing Disulfide Bonds. The repair process of self-healing polyurethane based on disulfide bonds is completed by the synergistic effect of disulfide bonds and hydrogen bonds. The molecular structure of polyurethane generally contains a large number of ester groups, ether bonds,

and urethane bonds. Therefore, a huge number of hydrogen bonds are formed in polyurethane materials and the existence of hydrogen bonds plays a vital role in the self-healing materials. The repair process of our designed self-healing polyurethane containing disulfide bonds mainly goes through two stages, as shown in Figure 6 (R stands for antifouling groups). The first stage is the intermolecular hydrogen-bond interaction on the coating surface, which makes the repaired surface tightly bonded to form an initial self-repair. The second stage is that the disulfide bonds on the repaired surface will form sulfur radicals centered on sulfur, which will attack other disulfide bonds for full exchange so as to achieve the final repair effect.

3.8. Lake Field Tests. We further evaluated the antifouling performance of polyurethane-based coatings through lake field experiments (Figure 7). After immersing in Taying Lake for one month, the blank panel without any coating was rusted, and the unmodified PU panel did not have good resistance to organisms as a result of the lack of antifouling groups. In contrast, the surfaces of the modified polyurethane-based coatings FPU, TPU, and FTPU were cleaner. After immersing for two months, the surface of the blank panel was rusted more severely, and there was adhesion of biofilms and shell organisms. The surfaces of the polyurethane-based coatings had different degrees of adhesion of silt, sand, and stone. These results indicated that the modified polyurethane-based coatings play an important role in preventing the adhesion of organisms. After three months, as the pollution pressure intensified, the surface of the blank panel was accumulated with shell organisms and a large amount of algae. In contrast, PU, FPU, TPU, and FTPU showed high efficiency when only a small amount of organisms was attached, indicating that the polyurethane-based coatings had certain anti-corrosion and static antifouling properties. In addition, the surfaces of FPU, TPU, and FTPU were almost clean after water jet treatment, and FTPU was the cleanest among them. It can be seen that the combination of fouling release and fouling resistance

**Figure 6.** Repair mechanism of polyurethane-based coatings containing disulfide bonds.

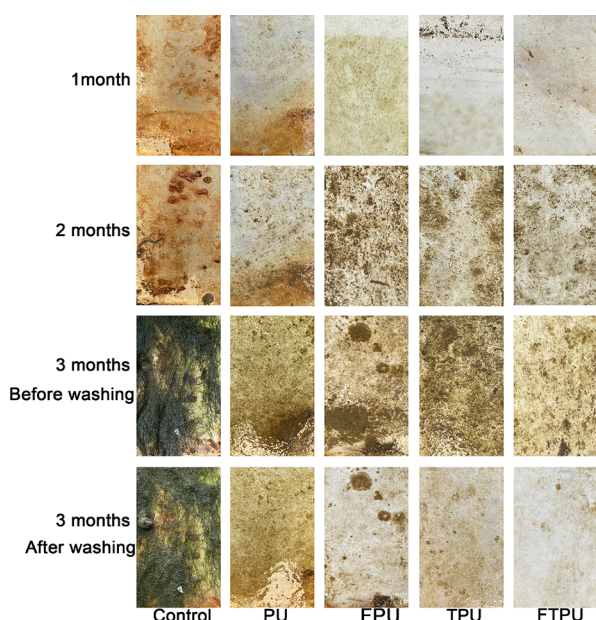


Figure 7. Pictures of polyurethane-based coatings after immersion in Taying Lake, Suzhou, China, before and after a water jet treatment (pressure: 0.4 MPa).

determines the long-term antifouling performance of grafted polyurethane-based coatings.

4. CONCLUSIONS

In this work, an environmentally friendly self-healing polyurethane antifouling coating was synthesized through the polymerization of IPDI, PTMEG, F-OH, TCSEA-OH, and S-S. The fouling release groups and the fouling resistance groups were uniformly fixed on the surface of the polyurethane coating in the form of side chains by chemical grafting to achieve a synergistic antifouling effect. The antifouling groups were not released into the external environment, so the coating was an environmentally friendly material. The coating could be self-repairing under thermal stimulation, room temperature, and water, effectively prolonging the antifouling life of the coating, which was mainly due to the synergistic self-repairing effect of the hydrogen bond and disulfide bond in the PU. The prepared cooperative antifouling coating (FTPU) had good substrate adhesion and could prevent the substrate from falling off. In addition, through lake-water antifouling experimental results, it could be seen that, in static and dynamic environments, compared with coatings (FPU and TPU) with only one type antifouling group added, cooperative antifouling coatings (FTPU) had better antifouling performance. Therefore, the prepared self-healing polyurethane-based antifouling paint (FTPU) had practical application potential in the field of lake antifouling.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsapm.2c01503>.

Water contact angle and surface energy test, adhesion tests, tensile performance characterization, and lake field tests (PDF)

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Notes

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